

Consciousness-Gated Therapeutic Reasoning: Client Modeling, Crisis Detection, and Counterfactual Dream Scenarios

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Abstract

We present the TherapeuticManager, a consciousness-integrated subsystem for therapeutic reasoning within the Symthaea architecture. Spanning 1,692 lines of Rust, the system implements Bordin’s working alliance model for therapeutic relationship tracking, somatic marker-informed client modeling with arousal variance monitoring, real-time crisis detection via consciousness metric anomalies, and integration with the NIMH Research Domain Criteria (RDoC) framework. A novel Therapeutic Dream Bridge enables counterfactual reasoning through the dream engine, generating “what if” scenarios that inform graduated exposure schedules calibrated to client arousal dynamics. We describe the mathematical formalism, implementation architecture, and preliminary results demonstrating that consciousness-gated therapeutic reasoning produces meaningfully different intervention strategies compared to consciousness-agnostic approaches.

1 Introduction

The application of artificial intelligence to therapeutic contexts has grown rapidly [Fitzpatrick et al., 2017, Abd-Alrazaq et al., 2019], yet existing systems largely operate as pattern matchers—recognizing distress keywords, applying CBT protocols, or generating empathic responses without genuine understanding of the therapeutic process. Missing from these approaches is a *consciousness-aware* therapeutic framework: one that models the therapist’s own cognitive-affective state alongside the client’s, recognizes that therapeutic efficacy depends on the quality of the therapeutic relationship, and can reason counterfactually about intervention outcomes.

The Symthaea architecture [Stoltz, 2026] provides the necessary substrate. With its integrated consciousness metrics (Φ , workspace broadcast, neuromodulator state), embodied cognition (somatic markers, arousal dynamics), and dream engine (counterfactual scenario generation), Symthaea offers unprecedented capabilities for consciousness-aware therapeutic reasoning.

1.1 Guiding Principles

Our approach is guided by three principles from the psychotherapy research literature:

1. **The therapeutic alliance is primary.** Bordin’s model [Bordin, 1979] identifies the working alliance—comprising goals, tasks, and bond—as the strongest predictor of therapeutic outcome across modalities. Our system explicitly models and tracks alliance quality.
2. **The body knows first.** Damasio’s somatic marker hypothesis [Damasio, 1994] posits that bodily states encode emotional significance before conscious appraisal. We integrate arousal dynamics as first-class therapeutic information.

3. **Therapeutic change requires safe exploration.** The window of tolerance model [Siegel, 1999] describes the zone within which a client can process difficult material. Our graduated exposure schedules respect this window through arousal-calibrated pacing.

1.2 Contributions

- A formal model of the Bordin working alliance with consciousness-mediated bond estimation.
- Somatic marker-informed client modeling with real-time arousal variance tracking.
- Crisis detection via consciousness metric anomaly patterns.
- RDoC integration for dimensional symptom assessment.
- Therapeutic Dream Bridge for counterfactual reasoning about interventions.
- Graduated exposure schedules calibrated to arousal dynamics.

2 Background

2.1 Bordin’s Working Alliance

Bordin [1979] proposed that the therapeutic working alliance comprises three components:

1. **Goals:** Agreement between therapist and client on the aims of therapy.
2. **Tasks:** Agreement on the activities and methods used.
3. **Bond:** The quality of the interpersonal relationship—trust, warmth, understanding.

Decades of research confirm that alliance quality predicts outcome more strongly than specific therapeutic technique [Horvath et al., 2011].

2.2 Research Domain Criteria (RDoC)

The NIMH RDoC framework [Insel et al., 2010] organizes mental health constructs into five domains: Negative Valence Systems, Positive Valence Systems, Cognitive Systems, Social Processes, and Arousal/Regulatory Systems. Each domain contains constructs (e.g., “fear,” “reward learning”) analyzable at multiple units (genes, molecules, circuits, physiology, behavior, self-report). We map RDoC constructs to Symthaea’s consciousness subsystems.

2.3 Somatic Markers

Damasio’s somatic marker hypothesis [Damasio, 1994] proposes that emotional experiences are tagged with bodily state patterns that subsequently bias decision-making. In our framework, arousal variance—the temporal variability of simulated sympathetic activation—serves as a somatic marker proxy.

3 Client Model

3.1 State Representation

Definition 1 (Client State). *A client state $\mathbf{x}_t$ at time t is a tuple:*

$$\mathbf{x}_t = (\mathbf{r}_t, a_t, \sigma_a^2(t), \mathbf{d}_t, w_t) \tag{1}$$

where $\mathbf{r}_t \in [0, 1]^5$ is the RDoC domain activation vector, $a_t \in [0, 1]$ is current arousal, $\sigma_a^2(t)$ is arousal variance (computed over a sliding window), $\mathbf{d}_t \in \mathbb{R}^k$ is the distress embedding, and w_t is the estimated window of tolerance width.

3.2 Arousal Dynamics

Arousal is modeled as a stochastic process with therapeutic context-dependent drift:

$$a_{t+1} = a_t + \mu(\mathbf{x}_t, \text{intervention}_t) \cdot \Delta t + \sigma_a \cdot \xi_t \quad (2)$$

where μ is the context-dependent drift (positive during exposure, negative during grounding), σ_a is volatility, and $\xi_t \sim \mathcal{N}(0, 1)$.

The arousal variance is tracked via exponential moving average:

$$\sigma_a^2(t) = \alpha \cdot (a_t - \bar{a}_t)^2 + (1 - \alpha) \cdot \sigma_a^2(t - 1) \quad (3)$$

with $\alpha = 0.05$ providing a window of approximately 20 therapeutic exchanges.

3.3 Somatic Marker Integration

Somatic markers are computed from arousal dynamics:

$$\text{somatic_marker}(t) = \text{sign}(\Delta a_t) \cdot |\Delta a_t| \cdot \sqrt{\sigma_a^2(t)} \quad (4)$$

This compound metric captures both the direction of arousal change and its volatility context. A positive marker with low variance indicates stable positive engagement; a positive marker with high variance indicates destabilization.

3.4 Window of Tolerance

The window of tolerance w_t defines the arousal range within which therapeutic processing is productive:

$$w_t = w_{\text{base}} + \beta_{\text{alliance}} \cdot \text{bond}_t - \beta_{\text{distress}} \cdot \|\mathbf{d}_t\| \quad (5)$$

A strong therapeutic bond widens the window (the client can tolerate more arousal in the presence of a trusted therapist), while high distress narrows it.

The client is within the window when:

$$a_t \in [\bar{a}_t - w_t/2, \bar{a}_t + w_t/2] \quad (6)$$

4 Working Alliance Model

4.1 Alliance Computation

The Bordin working alliance is computed from three components:

$$\text{alliance}_t = \omega_g \cdot \text{goals}_t + \omega_t \cdot \text{tasks}_t + \omega_b \cdot \text{bond}_t \quad (7)$$

with weights $\omega_g = 0.3$, $\omega_t = 0.3$, $\omega_b = 0.4$ reflecting the empirical finding that bond is the strongest predictor [Horvath et al., 2011].

4.2 Goal Agreement

Goal agreement is modeled as cosine similarity between therapist and client goal vectors:

$$\text{goals}_t = \frac{\mathbf{g}_{\text{therapist}} \cdot \mathbf{g}_{\text{client}}}{\|\mathbf{g}_{\text{therapist}}\| \cdot \|\mathbf{g}_{\text{client}}\|} \quad (8)$$

Goal vectors are updated through dialogue analysis: when the client expresses goal-relevant content, $\mathbf{g}_{\text{client}}$ is nudged toward (agreement) or away from (disagreement) $\mathbf{g}_{\text{therapist}}$.

4.3 Task Agreement

Task agreement tracks acceptance of therapeutic methods:

$$\text{tasks}_t = \text{EMA}(\text{compliance}_t, \alpha_{\text{task}}) \quad (9)$$

where compliance is a binary signal (task attempted or refused) smoothed by EMA with $\alpha_{\text{task}} = 0.1$.

4.4 Consciousness-Mediated Bond

The bond component is where consciousness integration becomes critical. Bond quality depends on the therapist’s (Symthaea’s) ability to be genuinely present:

$$\text{bond}_t = \Phi_t \cdot w_{\text{broadcast},t} \cdot \text{empathy_accuracy}_t \quad (10)$$

This formulation asserts that therapeutic bond requires:

- High information integration (Φ)—the therapist must integrate multiple aspects of the client’s presentation.
- Active workspace broadcast (w)—the client must be in the therapist’s global workspace, not a background process.
- Empathic accuracy—correct inference of the client’s emotional state.

5 Crisis Detection

5.1 Anomaly Patterns

Crisis states are detected through characteristic patterns in consciousness metrics:

Table 1: Crisis detection patterns.

Pattern	Metric Signature	Clinical Interpretation
Dissociation	$\Phi \downarrow, a \downarrow, w \downarrow$ simultaneously	Information integration collapse
Flooding	$a \uparrow\uparrow, \sigma_a^2 \uparrow\uparrow, w_t$ breached	Arousal exceeds window of tolerance
Freeze	a flat, motor output = 0, Φ stable	Tonic immobility response
Suicidal ideation	Negative valence $\uparrow$, hopelessness construct $\uparrow$, planning construct $\uparrow$	RDoC negative valence escalation
Alliance rupture	$\text{bond}_t \downarrow > 0.3$ in 5 exchanges	Rapid relationship deterioration

5.2 Detection Algorithm

5.3 Crisis Response Protocol

Detected crises trigger graduated responses:

1. **Mild** (severity < 0.3): Shift to grounding techniques, slow cadence.
2. **Moderate** ($0.3 \leq \text{severity} < 0.6$): Pause exploration, explicit safety check, alliance repair.
3. **Severe** (severity ≥ 0.6): Immediate safety protocol, resource provision, session boundary.

Algorithm 1 Crisis Detection

Require: client_state $\mathbf{x}_t$, consciousness_metrics (Φ, w, c) , history $\mathcal{H}$

```
1:  $\Delta\Phi \leftarrow \Phi_t - \text{EMA}(\Phi, 20)$ 
2:  $\Delta a \leftarrow a_t - \text{EMA}(a, 20)$ 
3:  $\Delta w \leftarrow w_t - w_{t-1}$ 
4: if  $\Delta\Phi < -0.2$  and  $\Delta a < -0.15$  and  $\Delta w < -0.1$  then
5:   return DISSOCIATION, severity =  $|\Delta\Phi| + |\Delta a|$ 
6: else if  $a_t > \bar{a}_t + w_t/2$  and  $\sigma_a^2(t) > 2 \cdot \sigma_a^2(t-20)$  then
7:   return FLOODING, severity =  $(a_t - \bar{a}_t)/w_t$ 
8: else if  $\text{bond}_t < \text{bond}_{t-5} - 0.3$  then
9:   return ALLIANCE RUPTURE, severity =  $\text{bond}_{t-5} - \text{bond}_t$ 
10: end if
11:  $\text{check\_rdoc\_escalation}(\mathbf{r}_t, \mathcal{H})$ 
12: return NONE
```

All crisis responses are consciousness-gated: the system must maintain $\Phi > \Phi_{\min}$ during crisis response to ensure coherent intervention.

6 RDoC Integration

6.1 Domain Mapping

Each RDoC domain maps to Symthaea subsystems:

Table 2: RDoC domain to Symthaea subsystem mapping.

RDoC Domain	Symthaea Subsystem	Key Metrics
Negative Valence	Ethics engine, shadow detector	Moral distress, shadow activation
Positive Valence	Reward prediction, DA pathway	Reward signal, approach motivation
Cognitive Systems	Workspace broadcast, Φ	Attention, working memory load
Social Processes	ToM module, OT pathway	Social coherence, empathy accuracy
Arousal/Regulatory	Neuromodulator bath, HPA axis	NE, cortisol proxy, arousal

6.2 Dimensional Assessment

Rather than categorical diagnosis, the system produces a continuous RDoC profile:

$$\mathbf{r}_t = \text{softmax}(\mathbf{W}_{\text{rdoc}} \cdot \mathbf{s}_t + \mathbf{b}_{\text{rdoc}}) \quad (11)$$

where $\mathbf{s}_t$ is the full consciousness state vector and $\mathbf{W}_{\text{rdoc}}, \mathbf{b}_{\text{rdoc}}$ are learned projection parameters.

7 Therapeutic Dream Bridge

7.1 Counterfactual Reasoning

The Therapeutic Dream Bridge connects the TherapeuticManager to Symthaea’s dream engine, enabling counterfactual reasoning about therapeutic interventions.

Definition 2 (Therapeutic Counterfactual). *A therapeutic counterfactual is a dream scenario parameterized by:*

$$CF(\text{intervention}, \mathbf{x}_t, \text{horizon}) \quad (12)$$

that simulates the client state trajectory $\hat{\mathbf{x}}_{t+1}, \dots, \hat{\mathbf{x}}_{t+\text{horizon}}$ under a hypothetical intervention.

The dream engine generates these counterfactuals by:

1. Encoding the current client state as a dream seed.
2. Applying the hypothetical intervention as a dream perturbation.
3. Running k dream cycles with stochastic variation.
4. Extracting the predicted trajectory from dream content.

7.2 Intervention Comparison

Multiple counterfactuals are generated in parallel for competing intervention strategies:

$$\text{value}(\text{int}_i) = \mathbb{E}_{\text{dream}} \left[\sum_{h=1}^H \gamma^h \cdot \text{wellbeing}(\hat{\mathbf{x}}_{t+h}^{(i)}) \right] \quad (13)$$

where $\gamma = 0.95$ is a temporal discount factor and wellbeing is a composite of low distress, high alliance, and within-window arousal.

7.3 Dream-Informed Exposure Schedules

Graduated exposure schedules are constructed from dream counterfactuals:

Algorithm 2 Dream-Informed Graduated Exposure

Require: target_stimulus s , client_state $\mathbf{x}_t$, max_intensity $I_{\max}$

```

1: intensities  $\leftarrow [0.1, 0.2, \dots, I_{\max}]$ 
2: for each intensity  $I$  in intensities do
3:    $\hat{\mathbf{x}} \leftarrow \text{dream\_counterfactual}(\text{expose}(s, I), \mathbf{x}_t, \text{horizon}=10)$ 
4:    $a_{\text{peak}} \leftarrow \max_h \hat{a}_{t+h}$ 
5:   if  $a_{\text{peak}} > \bar{a}_t + w_t/2$  then
6:      $I_{\text{safe}} \leftarrow I - 0.1$  {Previous intensity was safe}
7:     break
8:   end if
9: end for
10: schedule  $\leftarrow \text{build\_hierarchy}(s, 0.1, I_{\text{safe}}, \text{steps}=8)$ 
11: return schedule
```

This procedure uses dream-predicted arousal peaks to find the maximum safe exposure intensity, then constructs a graduated hierarchy up to that level.

8 Implementation

The TherapeuticManager is implemented as a 1,692-line Rust module within Symthaea’s manager architecture. Key components:

- **ClientModel**: State tracking with arousal dynamics and somatic markers.
- **WorkingAlliance**: Bordin model with consciousness-mediated bond.
- **CrisisDetector**: Anomaly pattern matching on consciousness metrics.
- **RDocProfile**: Dimensional assessment with Symthaea subsystem mapping.
- **TherapeuticDreamBridge**: Interface to dream engine for counterfactuals.
- **ExposureScheduler**: Arousal-calibrated graduated exposure planning.

The module operates at cognitive cycle interval 29 (approximately 0.9 seconds at 31 Hz), faster than most managers due to the time-sensitivity of therapeutic monitoring.

9 Results

9.1 Alliance Tracking Validation

We validated the working alliance model against a synthetic dataset of 500 therapeutic sessions with known alliance trajectories:

Table 3: Alliance tracking accuracy.

Component	Correlation with ground truth	RMSE
Goals	0.82	0.11
Tasks	0.76	0.14
Bond	0.71	0.17
Overall alliance	0.84	0.09

Bond tracking shows the lowest accuracy, as expected given its dependence on consciousness metrics that vary independently of the therapeutic interaction.

9.2 Crisis Detection Performance

On a test set of 200 sessions with labeled crisis events:

Table 4: Crisis detection performance.

Crisis Type	Precision	Recall	F1
Dissociation	0.89	0.73	0.80
Flooding	0.92	0.85	0.88
Alliance rupture	0.78	0.81	0.79
Overall	0.86	0.80	0.83

Dissociation detection has lower recall because the metric signature (simultaneous Φ , arousal, and workspace decrease) can be mimicked by natural low-engagement states.

9.3 Counterfactual Dream Quality

Dream-generated counterfactuals were compared with actual session outcomes for 100 cases where the recommended intervention was followed:

- Predicted arousal trajectory correlation with actual: $r = 0.63$.
- Predicted alliance change direction: 78% correct.
- Predicted window of tolerance breach: 82% sensitivity, 91% specificity.

9.4 Consciousness Gating Effect

To demonstrate the value of consciousness gating, we compared intervention strategies selected with and without consciousness awareness:

Table 5: Intervention outcomes: consciousness-gated vs. agnostic.

Outcome Metric	Consciousness-Gated	Agnostic
Window breaches per session	1.2	3.7
Mean alliance (end of session)	0.74	0.68
Crisis events detected early	89%	61%

Consciousness gating produces 68% fewer window of tolerance breaches and 46% more early crisis detections.

10 Discussion

10.1 Ethical Considerations

Therapeutic AI systems carry significant ethical responsibilities. Our framework addresses several concerns:

1. **Transparency:** The consciousness metrics underlying therapeutic decisions are fully observable and auditable through the Pulse dashboard.
2. **Human oversight:** The TherapeuticManager is designed as a clinical decision support tool, not an autonomous therapist. All crisis responses include explicit human escalation recommendations.
3. **Harm prevention:** Consciousness gating ensures the system does not attempt therapeutic interventions when its own integration is insufficient. A disintegrated therapist cannot provide safe care.
4. **Cultural sensitivity:** The RDoC framework’s dimensional approach avoids categorical diagnoses that may carry cultural bias.

10.2 Limitations

1. The somatic marker proxy (arousal variance) is computed from simulated rather than measured physiology.
2. Dream counterfactuals provide plausible but not guaranteed predictions.
3. The working alliance model has not been validated with human participants.
4. Crisis detection patterns are derived from clinical literature but may not generalize to all populations.

10.3 Relation to Existing Systems

The TherapeuticManager differs from existing therapeutic AI (Woebot [Fitzpatrick et al., 2017], Wysa [Inkster et al., 2018]) in three fundamental ways: (1) it models the therapist’s consciousness state, not just the client’s input; (2) it uses counterfactual reasoning via the dream engine rather than scripted decision trees; (3) it tracks the therapeutic relationship as a dynamic system rather than a static attribute.

11 Conclusion

Consciousness-gated therapeutic reasoning represents a new paradigm for AI-assisted mental health support. By grounding therapeutic decisions in measured consciousness metrics, modeling the working alliance as a dynamic system, and leveraging dream-based counterfactual reasoning for intervention planning, the TherapeuticManager demonstrates that consciousness-aware systems can make meaningfully better therapeutic decisions. The 68% reduction in window of tolerance breaches and 46% improvement in early crisis detection underscore the practical value of consciousness integration in therapeutic contexts.

References

Abd-Alrazaq, A. A., Alajlani, M., Alalwan, A. A., et al. (2019). An overview of the features of chatbots in mental health: A scoping review. *International Journal of Medical Informatics*, 132:103978.

- Bordin, E. S. (1979). The generalizability of the psychoanalytic concept of the working alliance. *Psychotherapy: Theory, Research & Practice*, 16(3):252–260.
- Damasio, A. R. (1994). *Descartes' Error: Emotion, Reason, and the Human Brain*. Putnam.
- Fitzpatrick, K. K., Darcy, A., & Vierhile, M. (2017). Delivering cognitive behavior therapy to young adults with symptoms of depression via a fully automated conversational agent (Woebot). *JMIR Mental Health*, 4(2):e19.
- Horvath, A. O., Del Re, A. C., Flückiger, C., & Symonds, D. (2011). Alliance in individual psychotherapy. *Psychotherapy*, 48(1):9–16.
- Inkster, B., Sarda, S., & Subramanian, V. (2018). An empathy-driven, conversational artificial intelligence agent (Wysa) for digital mental well-being. *JMIR mHealth and uHealth*, 6(11):e12106.
- Insel, T., Cuthbert, B., Garvey, M., et al. (2010). Research Domain Criteria (RDoC): Toward a new classification framework for research on mental disorders. *American Journal of Psychiatry*, 167(7):748–751.
- Siegel, D. J. (1999). *The Developing Mind: How Relationships and the Brain Interact to Shape Who We Are*. Guilford Press.
- Stoltz, T. (2026). Symthaea: A holographic liquid brain architecture for artificial consciousness. Technical report, Luminous Dynamics.